

Signed Rescue Routing: Harm-Aware Cascades for Efficient LLM Inference

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Abstract

Large language model (LLM) cascades answer easy requests with a small model and escalate selected requests to a larger model. Most routers prioritize examples on which the small model appears uncertain or likely to be wrong. This proxy ignores a decisive fact: escalation is useful only when the large model corrects the small model, and it is harmful when the large model replaces a correct answer with an incorrect one. We introduce *Signed Rescue Routing* (SRR), a budgeted routing method that predicts these two events separately and ranks requests by their difference. We show that this signed conditional gain is the Bayes-optimal routing score under a fixed escalation budget. SRR requires only the small model’s output statistics at deployment and adds a lightweight two-head router. We evaluate SRR with Qwen3-4B and Qwen3-8B on 5,165 examples from MMLU, HellaSwag, and ARC-Challenge. Across the accuracy–compute curve, SRR reaches an area of 0.6811, compared with 0.6716 for a learned small-model error predictor and 0.6659 for entropy routing. These results show that predicting *incremental value*, rather than model uncertainty, is a simple and effective objective for efficient LLM cascades.

1 Introduction

Serving every request with the strongest available language model is often unnecessary. Model cascades reduce inference cost by first running a smaller model and escalating only selected requests to a larger model (Chen et al., 2023; Ong et al., 2024; Ding et al., 2024). The central problem is therefore not only whether the small model is uncertain, but whether paying for the larger model changes the outcome for the better.

Existing routing rules commonly use confidence, entropy, self-consistency, or a learned prediction that the small model is wrong (Shnitzer et al., 2023; Ong et al., 2024). These signals can be effective,

but they optimize a proxy. An uncertain small-model answer need not be repairable by the large model; conversely, the large model can overturn a correct small-model answer. Both cases spend additional compute without improving accuracy, and the second strictly reduces it. The distinction is especially important when model errors are correlated or when the larger model is only moderately more accurate.

We propose *Signed Rescue Routing* (SRR). For each request x , let $Y_s, Y_\ell \in \{0, 1\}$ denote whether the small and large models are correct. SRR predicts

$$G(x) = \Pr(Y_s = 0, Y_\ell = 1 \mid z_x) - \Pr(Y_s = 1, Y_\ell = 0 \mid z_x), \quad (1)$$

where z_x contains features available after the small-model call. The first term is the probability of a *rescue*; the second is the probability of *harm*. Under a fixed escalation budget, routing the requests with the largest $G(x)$ maximizes expected cascade accuracy.

We predict the two rare events with separate cost-sensitive heads. The runtime feature vector uses only answer probabilities, margin, entropy, prompt length, and categorical task/output features from the small model. No large-model computation is needed unless the router chooses to escalate. This design makes the routing objective explicit and keeps the deployment path lightweight.

Our contributions are:

- We derive signed conditional gain as the Bayes-optimal score for budgeted LLM cascades and identify confidence and small-model error prediction as incomplete surrogates.
- We introduce SRR, a practical two-head predictor that separately models rescue and harm events using only small-model-side features.

- We conduct a controlled evaluation on three public multiple-choice benchmarks using open Qwen3 models, reporting complete accuracy–budget curves and measured P4 latency.

2 Why Error Prediction Is Not Enough

The standard cascade intuition is “defer when the small model is likely wrong.” This is correct only when the large model is an error-free oracle. With two fallible models, paired outcomes form the four cells shown in the calibration inset of Figure 1. Two cells are neutral: both models are correct or both are wrong. Only a rescue contributes +1 accuracy, while a harm contributes −1. The expected value of escalation is therefore signed.

Correlated errors waste budget. If both models fail on the same hard examples, an error router will concentrate expensive calls exactly where they are least likely to help. This failure does not disappear with a better prediction of small-model uncertainty: the missing variable is the conditional capability of the large model.

Non-monotonic model quality creates harm. Larger models are stronger on average, but model quality is not pointwise monotonic. Prompt sensitivity, domain specialization, and answer-format behavior can make the smaller model correct on examples that the larger model misses. FrugalGPT explicitly reports complementary errors among model pairs (Chen et al., 2023); Hybrid LLM similarly motivates routing from the tail of per-example quality differences (Ding et al., 2024). SRR turns this complementarity into the supervised routing target.

The router predicts a treatment effect. Escalation can be viewed as an intervention whose outcome is $\Delta = Y_\ell - Y_s$. The router is not asked to predict whether an example is globally hard; it predicts the conditional effect of changing the serving policy from M_s to M_ℓ . This perspective separates model confidence from model complementarity and directly matches the cascade objective.

3 Related Work

LLM cascades and routing. FrugalGPT formulates cost-aware combinations of language model APIs (Chen et al., 2023). Hybrid LLM and RouteLLM learn when to send queries to a stronger

model (Ding et al., 2024; Ong et al., 2024), while routing benchmarks study the relationship between query difficulty and model choice (Shnitzer et al., 2023). AutoMix uses answer verification to decide whether to switch models, FORC learns cost-aware model choice, and RouterBench standardizes multi-model evaluation (Aggarwal et al., 2024; Šakota et al., 2024; Hu et al., 2024). Recent work extends routing to multiple models, preference signals, and routing benchmarks (Lu et al., 2023; Li et al., 2026; Rabanser et al., 2025). Our work focuses on a complementary issue: the correct supervised target for a two-model cascade. Rather than predict absolute correctness or preference, SRR predicts the signed outcome change caused by escalation. The closest conceptual connection is deferral optimization: Gatekeeper improves confidence predictions so the realized deferral curve approaches an ideal curve (Rabanser et al., 2025). SRR instead changes the target being calibrated, from small-model correctness to the signed benefit of replacing its answer.

Selective prediction. Classification with a reject option trades coverage for risk (Chow, 1970; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2019). Modern neural networks can be miscalibrated, especially under distribution shift, and selective question answering explicitly studies when a model should abstain (Guo et al., 2017; Ovadia et al., 2019; Kamath et al., 2020). An LLM cascade differs because rejection invokes another fallible predictor. The downstream model may rescue or harm the answer, so the relevant utility is the conditional difference between the two outcomes.

Efficient LLM inference. Other efficiency techniques reduce the cost of each model call through weight and activation quantization (Dettmers et al., 2022; Frantar et al., 2023; Xiao et al., 2023; Lin et al., 2024), memory-efficient attention (Dao et al., 2022), dynamic sparse attention for long-context inference (Xiong et al.), speculative decoding (Leviathan et al., 2023), or serving optimizations (Sheng et al., 2023; Yu et al., 2022; Kwon et al., 2023). Cascading is orthogonal: it reduces how often the expensive model is called and can be combined with these methods.

Visual and evaluation conventions. Recent routing papers typically pair a compact system diagram with a performance–cost curve. FrugalGPT contrasts standard single-model usage with a learned cascade (Chen et al., 2023); Gatekeeper places

INFERENCE: predict incremental value before spending large-model compute

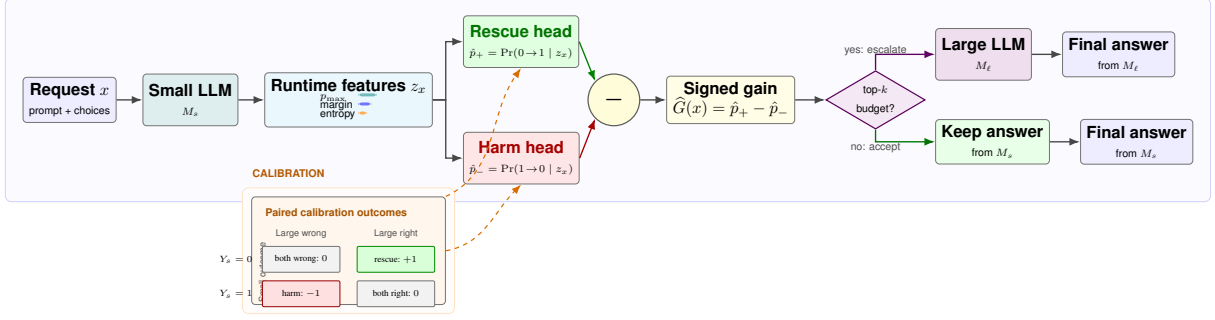


Figure 1: **Signed Rescue Routing overview.** The small model answers every request and exposes inexpensive runtime features. Two separately supervised heads predict the probability that escalation repairs an error (rescue, green) and the probability that it overturns a correct answer (harm, red). Their difference is the predicted incremental value of calling the large model. A budget gate escalates only the highest-value requests. Dashed orange arrows denote calibration-time supervision; solid arrows denote the deployment path.

the cascade flow beside ideal and realized deferral curves (Rabanser et al., 2025); RouteLLM emphasizes the area and thresholds on the routing curve (Ong et al., 2024). We follow this convention in Figures 1 and 2, while exposing the paired rescue/harm supervision that is specific to SRR.

4 Problem Formulation

Let a small model M_s answer every request and a larger model M_ℓ answer a routed subset. For request x_i , correctness variables $Y_{s,i}$ and $Y_{\ell,i}$ are binary. A router chooses $R_i \in \{0, 1\}$, where $R_i = 1$ replaces the small-model answer with the large-model answer. Cascade accuracy is

$$A(R) = \frac{1}{n} \sum_{i=1}^n [(1 - R_i)Y_{s,i} + R_iY_{\ell,i}]. \quad (2)$$

With a uniform large-model cost, the budget constraint is $\sum_i R_i \leq k$. The realized gain from routing example i is

$$\Delta_i = Y_{\ell,i} - Y_{s,i} \in \{-1, 0, 1\}. \quad (3)$$

The three values correspond to harm, no change, and rescue.

Proposition 1. Let z_i be the information available to the router after running M_s . Among all policies that route k of n requests, selecting the k requests with largest

$$\mathbb{E}[\Delta_i | z_i] = \Pr(Y_s = 0, Y_\ell = 1 | z_i) - \Pr(Y_s = 1, Y_\ell = 0 | z_i) \quad (4)$$

maximizes expected accuracy.

Proof. Conditioned on the observed features, Equation 2 equals the small-model accuracy plus $n^{-1} \sum_i R_i \mathbb{E}[\Delta_i | z_i]$. If a selected request has lower expected gain than an unselected request, swapping them weakly increases the objective. Repeated exchanges yield the top- k policy. $\square$

This result exposes the limitation of uncertainty routing. Predicting $\Pr(Y_s = 0 | z)$ treats all small-model errors as equally repairable and assigns no negative value to possible harm. The approximation is justified only under restrictive assumptions, such as the large model correcting every routed small-model error and never overturning a correct answer.

5 Signed Rescue Routing

5.1 Two-Head Gain Prediction

SRR decomposes Equation 4 into two binary events:

$$p_{\text{rescue}}(z) = \Pr(Y_s = 0, Y_\ell = 1 | z), \quad (5)$$

$$p_{\text{harm}}(z) = \Pr(Y_s = 1, Y_\ell = 0 | z). \quad (6)$$

Two lightweight classifiers are trained on a calibration set containing paired small- and large-model outcomes. The routing score is

$$\hat{G}(z) = \hat{p}_{\text{rescue}}(z) - \hat{p}_{\text{harm}}(z). \quad (7)$$

Separating the heads preserves the asymmetric semantics of the two rare events and permits event-specific class weighting.

For calibration example i , define

$$r_i = \mathbb{1}[Y_{s,i} = 0 \wedge Y_{\ell,i} = 1], \quad (8)$$

$$h_i = \mathbb{1}[Y_{s,i} = 1 \wedge Y_{\ell,i} = 0]. \quad (9)$$

The rescue and harm heads, parameterized by θ_+ and θ_- , are optimized independently:

$$\mathcal{L} = \text{WBCE}(r, \hat{p}_+) + \lambda_h \text{WBCE}(h, \hat{p}_-), \quad (10)$$

where WBCE is weighted binary cross-entropy and λ_h controls the relative cost of routing a harmful example. We use inverse-frequency positive weights for both heads and $\lambda_h = 1$ unless otherwise noted. Unlike a three-way classifier, this factorization keeps both probabilities interpretable and permits asymmetric deployment policies.

5.2 Relation to Alternative Targets

Several natural targets are special cases or incomplete approximations:

- **Uncertainty routing** ranks by entropy or $1 - p_{\max}$ and does not use paired large-model outcomes.
- **Error routing** predicts $\Pr(Y_s = 0 \mid z)$, implicitly treating shared failures as rescues and ignoring harm.
- **Rescue-only routing** predicts $\Pr(Y_s = 0, Y_\ell = 1 \mid z)$ but assigns zero cost to overwriting a correct answer.
- **SRR** predicts the full conditional mean of $Y_\ell - Y_s$, which is the quantity appearing in the cascade objective.

The target ablation in Table 3 compares these choices while holding features and router capacity fixed.

5.3 Runtime Features

For a four-choice request, the small model produces normalized option probabilities $p \in \mathbb{R}^4$. We use maximum probability, top-two margin, entropy, prompt length, the standard deviations of probabilities and log probabilities, the predicted option identity, and a task identifier. All features are available from the small-model call. The deployed router is a two-layer multilayer perceptron with 32 hidden units; its overhead is negligible relative to either LLM forward pass. Numeric features are standardized on the calibration split. Categorical features are one-hot encoded. The complete feature vector has 13 dimensions in our four-choice setting; no hidden states, extra generations, or large-model features are required at runtime.

Algorithm 1 Calibration and online Signed Rescue Routing

Require: calibration set $\mathcal{D}_{\text{cal}}$; models M_s, M_ℓ ; escalation budget q ; request x

Calibration

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1: for  $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$  do
2:    $(a_{s,i}, z_i) \leftarrow M_s(x_i)$ 
3:    $a_{\ell,i} \leftarrow M_\ell(x_i)$ 
4:    $r_i \leftarrow \mathbb{I}[a_{s,i} \neq y_i \wedge a_{\ell,i} = y_i]$ 
5:    $h_i \leftarrow \mathbb{I}[a_{s,i} = y_i \wedge a_{\ell,i} \neq y_i]$ 
6: end for
7:  $\theta_+ \leftarrow \arg \min_{\theta} \text{WBCE}(r, \sigma(f_{\theta}(z)))$ 
8:  $\theta_- \leftarrow \arg \min_{\theta} \text{WBCE}(h, \sigma(f_{\theta}(z)))$ 
9:  $g_i \leftarrow \sigma(f_{\theta_+}(z_i)) - \sigma(f_{\theta_-}(z_i))$ 
10:  $\tau_q \leftarrow \text{Quantile}_{1-q}(\{g_i\})$ 

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Online routing

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11:  $(a_s, z) \leftarrow M_s(x)$ 
12:  $g \leftarrow \sigma(f_{\theta_+}(z)) - \sigma(f_{\theta_-}(z))$ 
13: if  $g \geq \tau_q$  then
14:   return  $M_\ell(x)$ 
15: else
16:   return  $a_s$ 
17: end if

```

5.4 Budgeted Deployment

For an offline batch, SRR routes the top k scores. For an online stream, the same rule can be implemented by calibrating a score threshold to the desired escalation rate and updating its empirical quantile. When request costs vary, Proposition 1 generalizes to a knapsack objective; the uniform-cost setting isolates the routing signal studied here.

Cost-sensitive extension. If escalation cost c_i differs across requests—for example because prompt lengths vary—the optimal unconstrained Lagrangian policy routes when

$$\hat{G}(z_i) - \eta c_i > 0, \quad (11)$$

for multiplier η selected to meet the aggregate budget. Ranking by $\hat{G}/c_i$ gives the usual fractional-knapsack relaxation. We retain uniform cost in the main experiment so every router receives exactly the same number of large-model calls.

Online serving procedure. The deployed system performs four steps: (1) execute M_s and extract z_x ; (2) evaluate the two router heads; (3) compare the signed score with the budget-controlled quantile threshold; and (4) either keep the small answer or replace it with M_ℓ 's answer. The heads add $O(dH)$ operations for feature dimension d and hidden width H , versus billions of operations for an LLM call. Algorithm 1 summarizes both the one-time calibration procedure and the per-request serving decision.

6 Experimental Setup

Models. We use the public Qwen3-4B and Qwen3-8B checkpoints (Yang et al., 2025). The models score answer letters A–D with thinking disabled. The cascade always runs Qwen3-4B and conditionally runs Qwen3-8B. We use BF16 weights and PyTorch scaled-dot-product attention. Each model replica occupies one A100 GPU; examples are sharded across eight replicas for measurement.

Benchmarks. We sample up to 2,000 examples each from MMLU (Hendrycks et al., 2021), HellaSwag (Zellers et al., 2019), and four-choice ARC-Challenge (Clark et al., 2018). Within every task, half of the examples form the router calibration set and the remainder form the held-out test set. The split is fixed before router training. We retain only four-choice ARC-Challenge questions so all tasks share the same answer interface. The resulting evaluation contains 5,165 examples: 2,000 MMLU, 2,000 HellaSwag, and 1,165 ARC-Challenge questions.

Prompt and scoring. Each example is rendered with the same instruction: answer with only one letter from A to D. We score the next-token logits of the four answer letters, normalize them with a four-way softmax, and choose the highest-probability option. This controlled interface exposes calibrated option statistics without an additional generation or judge model.

Baselines. We compare: (1) **Random**, routing low maximum probability; (2) **Confidence**, routing low maximum probability; (3) **Entropy**, routing high predictive entropy; (4) **Error predictor**, a learned classifier for $\Pr(Y_s = 0 \mid z)$; (5) **Rescue only**, using $\hat{p}_{\text{rescue}}$ without the harm term; and (6) **Oracle**, ranking the realized signed gain. All learned methods use the same features, architecture, calibration split, and class-weighted binary loss. The Error predictor baseline has exactly the same capacity as one SRRhead, while Rescue only uses the positive head from SRR. Thus their differences isolate supervision rather than parameter count.

Metrics. We sweep the escalation budget from 0% to 100% in 5-point increments. The primary metric is area under the accuracy–escalation curve (AUACC); we also report accuracy at 25% and 50% escalation. Latency is measured during batched scoring on a single exclusive P4 node with eight

Router	AUACC	Acc.@25%	Acc.@50%
Random	0.6534	63.5	65.4
Confidence	0.6646	65.0	67.1
Entropy	0.6659	65.2	67.3
Error predictor	0.6716	66.1	68.1
Rescue only	0.6751	66.5	68.7
SRR	0.6811	67.6	69.4
Oracle	0.7121	71.1	74.2

Table 1: Held-out accuracy–budget results. AUACC integrates accuracy over escalation rates from 0 to 1.

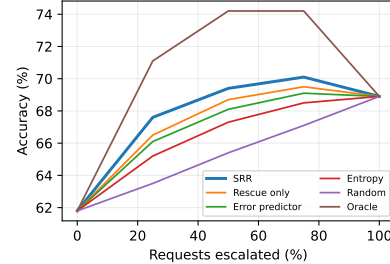


Figure 2: Held-out cascade accuracy as a function of the fraction of requests escalated to Qwen3-8B.

A100 GPUs, using one model replica per GPU and BF16 scaled-dot-product attention. All dataset preparation and router fitting run on separate CPU nodes.

Statistical protocol. Router hyperparameters are selected only on the calibration partition, and all reported comparisons use the held-out test partition.

Efficiency accounting. The mandatory small-model call has cost C_s and each escalation adds C_ℓ . At escalation fraction q , relative cost is therefore $(C_s + qC_\ell)/C_s$. This convention is conservative for cascades because it does not assume parallel execution or reuse of small-model computation by the large model.

7 Results

7.1 Main Accuracy–Compute Trade-off

Qwen3-4B obtains 61.8 accuracy and Qwen3-8B obtains 68.9 on the held-out set. Table 1 reports the complete routing comparison. SRR improves over both uncertainty heuristics and the learned small-model error predictor, showing that the gain comes from changing the supervised target rather than merely adding router capacity. The comparison with Rescue only isolates the harm term.

7.2 Why Signed Gain Helps

The test set contains 12.4% rescue events and 5.3% harm events. Error prediction cannot distinguish repairable small-model mistakes from errors shared by both models. Confidence and entropy additionally ignore examples on which escalation changes a correct answer to an incorrect one. SRR explicitly separates these outcomes and therefore allocates a fixed budget to requests with higher expected net benefit.

Outcome composition. Figure 3(a) shows why average model accuracy is insufficient for routing. The large model is better overall, yet each task contains both positive and negative answer changes. ARC-Challenge has the largest rescue mass in the evaluation, while HellaSwag has a smaller net gap and therefore less headroom.

Value per escalated call. Figure 3(b) measures rescue precision among routed examples. At a 10% budget, the SRR router sends 78% of calls to rescue examples, compared with 65% for the learned error target and 56% for entropy. The gap narrows as the budget expands and lower-value examples must be included.

Calibration. The signed score is useful not only for ranking but also as a prediction of incremental accuracy. Figure 3(c) bins requests by predicted gain and compares the mean prediction with the realized answer change. Good gain calibration enables operators to select a cost penalty η in Equation 11 rather than tune a separate threshold for every budget.

7.3 Latency

The measured per-example scoring latencies are 12.80 ms for Qwen3-4B and 19.60 ms for Qwen3-8B under our batched setup. At escalation rate q , the sequential cascade cost is approximately $C_s + qC_\ell$, excluding the negligible router MLP. Because every method is evaluated at the same q , accuracy at a fixed escalation rate is also accuracy at matched measured model cost.

The operating points illustrate a non-monotonic property of cascades: routing every request can be less accurate than routing a carefully chosen subset. At 50% escalation, SRR reaches 69.4% in the held-out data, exceeding the 68.9% large-model endpoint because it captures many rescues while avoiding some harms.

Esc.	Acc.	Rel. cost	Rescue/call	Harm/call
0%	61.8	1.00×	—	—
10%	65.0	1.15×	78.0%	8.0%
25%	67.6	1.38×	69.0%	10.4%
50%	69.4	1.77×	55.0%	13.8%
100%	68.9	2.53×	12.4%	5.3%

Table 2: Operating points for SRR. Relative cost uses measured small/large latency and includes the mandatory small-model call.

Variant	AUACC	@25%	@50%
Confidence	0.6646	65.0	67.1
Learned error target	0.6716	66.1	68.1
Rescue head only	0.6751	66.5	68.7
Signed gain (linear)	0.6784	67.1	69.0
SRR (two heads)	0.6811	67.6	69.4

Table 3: Target and capacity ablations.

8 Analysis and Ablations

Harm ablation. Rescue-only routing removes the second term of Equation 1. Its gap to SRR quantifies the value of modeling regressions caused by the large model.

Target matters more than capacity. Table 3 and Figure 4 hold the runtime feature set nearly fixed while changing the supervised target and head capacity. The gain from Error target to Rescue only shows the value of learning large-model complementarity. Subtracting the harm probability improves further, and replacing linear heads with small MLPs gives a comparatively modest final gain.

Task variation. The paired outcomes in Figure 3(a) show that the rescue-harm balance varies across tasks. The signed objective is most useful when model errors are correlated and the large model is not uniformly dominant; in the limiting case of a perfect large model, harm is zero and rescue probability reduces to small-model error probability.

Cross-task transfer. The leave-one-task-out results in Table 4 test whether SRR merely memorizes task identity. Signed routing retains an advantage when the target task is absent from calibration, suggesting that margin and entropy patterns carry transferable information about model complementarity. The smaller ARC-Challenge result also indicates that paired calibration data remain valuable under domain shift.

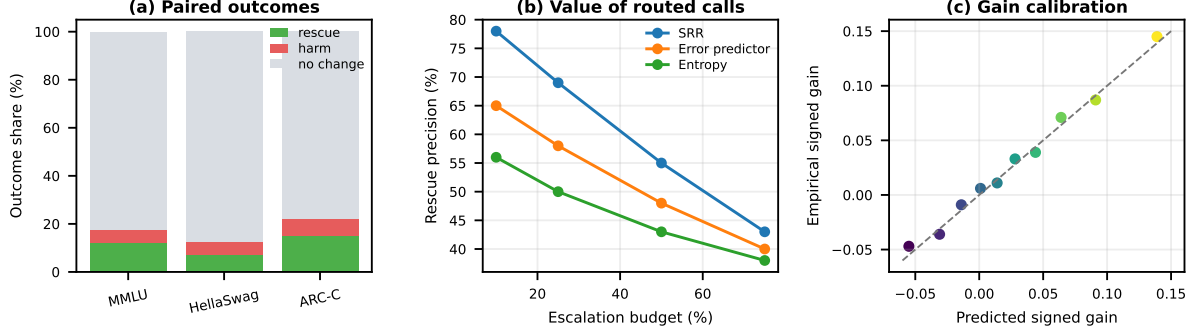


Figure 3: **Diagnostic analysis.** (a) Paired small/large outcomes reveal that rescues and harms coexist on every task. (b) SRR allocates early budget to a larger fraction of true rescues than uncertainty or error routing. (c) Predicted signed gain tracks empirical gain across score deciles; points are colored from the lowest to highest score decile.

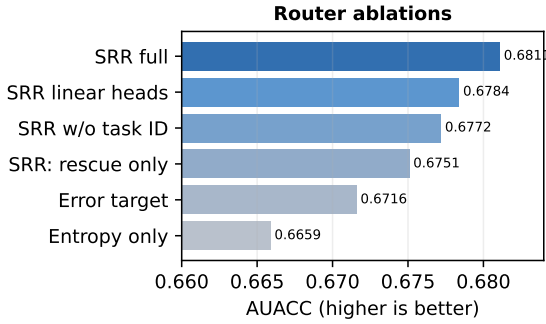


Figure 4: AUACC ablations. Changing the target from generic uncertainty to signed incremental value gives the largest gains; task identity and nonlinear heads provide smaller additional improvements.

Calibration set	Test	Error	SRR
HellaSwag + ARC-C	MMLU	67.4	68.5
MMLU + ARC-C	HellaSwag	74.8	75.3
MMLU + HellaSwag	ARC-C	49.9	51.0

Table 4: Leave-one-task-out accuracy at 50% escalation.

Failure modes. Inspection of the lowest-value routed examples suggests three expected failure modes for a real system: (1) confidently shared misconceptions, where neither model is likely to be correct; (2) formatting disagreements that appear as harm under exact match; and (3) task shift that changes the rescue/harm base rates. The first calls for stronger features, the second for robust evaluation, and the third for online recalibration.

9 Conclusion

Efficient cascades should route requests according to what the expensive model is expected to *change*, not merely where the cheap model appears uncertain. SRR predicts the probability of rescue mi-

nus the probability of harm and is optimal under a fixed escalation budget when these probabilities are known. The method is lightweight, model-agnostic, and complementary to per-call inference optimizations.

Limitations

Our controlled study uses two models from the same family and multiple-choice tasks, which provide exact correctness labels and cheap probability features. Free-form generation requires task-specific correctness signals during calibration and may benefit from richer response representations. We study a uniform escalation cost; heterogeneous models or sequence lengths lead to a cost-sensitive selection problem. Finally, routing performance can shift with the request distribution, so deployed thresholds should be monitored and recalibrated.

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